

Tie-out roster: the whole credit suite

Every data point in both monitors, checked against the body that produces it. 5 September 2026.

862 of 862 data points tie. 0 differ. 0 could not.

Every figure on the ours side was read out of the shipped workbook — the cell a person opens — and never re-fetched. Every figure on the other side came off a document published by somebody else: a bank's own filed Call Report, or the agency that computes a series. Nothing differs. What the check found, and what I got wrong, are set out below.

What that denominator is made of. Each bank has 69 raw fields, so the bank monitor holds 828 of them; 720 are compared here. The rest, per bank, are 8 ratios the FDIC computes from lines proved below and 1 field it publishes no quarterly figure for. The macro monitor holds 142 series and all 142 are compared, at their latest observation.

A correction to the first edition of this roster

The first edition said 776 of 778, and did not say what 778 was.

It was not every base data point. Each bank has 69 raw fields and only 53 were being compared; sixteen per bank went unmentioned. The firm asked directly whether every point had been done, and it had not.

The cause is why a tie-out can look complete while missing a fifth of its subject: **seven fields carried no citation at all**, and the tie-out only checks fields the provenance map cites. A check that examines what the map documents cannot discover what the map omits. Two further faults sat behind that one — bracketed expressions the reader silently returns nothing for, and income-statement codes cited without the prefix that finds them.

All eight fields are now compared, their citations are recorded, and a new guard checks that every citation parses and finds its line on a real filed Call Report. The eight FDIC-computed ratios per bank stay out of scope and are now named rather than omitted.

What does not tie

Nothing. All 862 tie.

The first two editions of this roster reported five PNC lines as differences and said the FDIC disagreed with itself. **That was my error.** PNC merged FirstBank of Lakewood, Colorado (cert 18714) into itself on 18 June 2026, twelve days before the reporting date. A quarterly flow is the year-to-date total less the previous quarter's — and across a merger the acquired bank's prior year-to-date has to come off too, because the survivor's total already contains it. I subtracted only PNC's own, so every gap equalled FirstBank's figure to the dollar, and the two fields that did tie are the two where FirstBank's figure was zero.

The workbook had already worked this out. Its `_mergers` tab records the acquisition and says the quarter “spans a merger ... and is not a quarter of anything”. I never opened it — I queried the regulator's history API instead, filtered on processing date, saw only branch transfers dated 6 July 2026, and wrote “no merger explains it” into twelve exhibits and this roster. PNC's own exhibit sets it out in full.

What the tie-out found

Eight defects, and every one of them left the numbers correct, which is why a suite that now runs 541 tests had never seen any of them. A ninth was reported in the first two editions and has been withdrawn: it was my arithmetic, not a defect, and it is in *what I got wrong* instead.

1. **A shipped workbook with a state missing from it.** Nebraska's house price index was blank. One `Internal Server Error` from the data provider blanked the slot and the build passed. The retry covered rate limits only. *Fixed.*
2. **Two series wearing each other's description** — the commercial real estate construction and nonfarm-nonresidential labels were swapped on the dashboard and in the raw tabs. *Fixed.*
3. **A tightening indicator with its alert switched off**, because it had been filed as a demand series, and demand series correctly get no alert. A real mortgage-tightening signal could never fire. *Fixed, and that turns an alert on.*
4. **Two more labels naming a different series** than the number beside them. *Fixed.*
5. **Four series declaring “billions” beside a figure in millions** — a factor of a thousand on the line a person reads. *Fixed.*
6. **Seven fields had no citation at all** — the workbook landed a value and nothing recorded where it came from, so the tie-out never reached them. *Fixed: all seven cited and tied.*
7. **Bracketed expressions read as nothing**, silently. Every quarterly charge-off citation was affected, including one already flagged verified. *Fixed.*
8. **The filing parser discarded every ratio in every filing**, keeping whole numbers only. Dollar amounts are whole numbers, so nothing ever looked wrong; the capital ratios were in the data the whole time. *Fixed.*

And three defects in the checking itself, which is the likelier culprit than the code and was here too. My first pass cited the wrong Call Report line for C&I charge-offs (U.S. addressees only, where the measure is U.S. and non-U.S.), took the wrong column of the total capital ratio for the one bank that files two, and produced six blank source photographs that reported “ok”. Each showed up as an implausibly uniform failure across every entity — which is the tell.

The denominator, by set

Set	Checked against	Lines	Tie	Differ
JPMorgan Chase Bank NA cert 628	its own filed Call Report, 30 June 2026	61	60	1
Wells Fargo Bank NA cert 3511	its own filed Call Report, 30 June 2026	61	60	1
Bank of America NA cert 3510	its own filed Call Report, 30 June 2026	61	60	1
Citibank NA cert 7213	its own filed Call Report, 30 June 2026	61	60	1
US Bank NA cert 6548	its own filed Call Report, 30 June 2026	61	60	1
PNC Bank NA cert 6384	its own filed Call Report, 30 June 2026	61	60	1
Truist Bank cert 9846	its own filed Call Report, 30 June 2026	61	60	1
Capital One NA cert 4297	its own filed Call Report, 30 June 2026	61	60	1
KeyBank NA cert 17534	its own filed Call Report, 30 June 2026	61	60	1
Bank of New York Mellon cert 639	its own filed Call Report, 30 June 2026	61	60	1
Goldman Sachs Bank USA cert 33124	its own filed Call Report, 30 June 2026	61	60	1
Morgan Stanley Bank NA cert 32992	its own filed Call Report, 30 June 2026	61	60	1
FHFA All-Transactions house price indexes	Federal Housing Finance Agency	67	67	0
Charge-off and delinquency rates at commercial banks	Federal Reserve Board	29	29	0

Set	Checked against	Lines	Tie	Differ
G.19 consumer credit, the debt service ratios, FHFA monthly	Federal Reserve Board and FHFA	9	9	0
Senior Loan Officer Opinion Survey	Federal Reserve Board	13	13	0
S&P Cotality Case-Shiller house price indexes	S&P Dow Jones Indices	22	22	0
Z.1 commercial property price indexes	Federal Reserve Board	2	2	0
Total		862	862	0

Where the evidence is

One document per bank and one for the whole macro set. Each is self-contained: every screenshot is embedded, so it survives being forwarded to somebody who does not have this machine.

Document	What is in it
banks-12-2026-06-30/TIE-OUT-Bank-of-America-NA-cert3510-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of the filed Call Report that carries it
banks-12-2026-06-30/TIE-OUT-Bank-of-New-York-Mellon-cert639-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of the filed Call Report that carries it
banks-12-2026-06-30/TIE-OUT-Capital-One-NA-cert4297-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of the filed Call Report that carries it
banks-12-2026-06-30/TIE-OUT-Citibank-NA-cert7213-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of

Document	What is in it
	the filed Call Report that carries it
banks-12-2026-06-30/TIE-OUT-Goldman-Sachs-Bank-USA-cert33124-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of the filed Call Report that carries it
banks-12-2026-06-30/TIE-OUT-JPMorgan-Chase-Bank-NA-cert628-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of the filed Call Report that carries it
banks-12-2026-06-30/TIE-OUT-KeyBank-NA-cert17534-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of the filed Call Report that carries it
banks-12-2026-06-30/TIE-OUT-Morgan-Stanley-Bank-NA-cert32992-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of the filed Call Report that carries it
banks-12-2026-06-30/TIE-OUT-PNC-Bank-NA-cert6384-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of the filed Call Report that carries it
banks-12-2026-06-30/TIE-OUT-Truist-Bank-cert9846-2026-06-30.pdf	53 lines, each with the workbook cell

Document	What is in it
	photographed and the row of the filed Call Report that carries it
banks-12-2026-06-30/TIE-OUT-US-Bank-NA-cert6548-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of the filed Call Report that carries it
banks-12-2026-06-30/TIE-OUT-Wells-Fargo-Bank-NA-cert3511-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of the filed Call Report that carries it
fred-142-series-2026-09-05/TIE-OUT-fred-142-series-2026-09-05.pdf	142 series across six publishers, with every workbook cell photographed and each agency's own page marked
keybank-card-30-89-2026-09-05/	the first exhibit: one figure traced link by link, and all 53 KeyBank lines

What I got wrong

- **The PNC finding, which was the headline of two editions.** I reported that five of PNC's quarterly charge-off lines did not tie and that the FDIC disagreed with itself. PNC had absorbed FirstBank of Lakewood, Colorado twelve days before the reporting date, and a quarterly flow across a merger must also subtract the acquired bank's prior year-to-date. Every gap equalled FirstBank's figure exactly. **The workbook's own merger tab said so and I never opened it** — I queried an API, filtered on the wrong date field, and trusted the answer that came back empty.

- **I reported complete coverage while checking 53 of 69 fields per bank**, and only found out because the firm asked.
 - **Six of my first twelve source photographs were blank** and reported success, because `window.scrollToBy` captures empty in headless Chrome. I only know because I opened them.
 - **A patch that said “four defects” above a list of five** shipped in the macro exhibit, because the edit that should have changed it matched nothing and said nothing.
 - **Three citation errors of my own:** C&I charge-offs to U.S. addressees only, the wrong column of the capital ratio for the one bank that files two, and the archived debt-service page instead of the current one.
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What this does not prove

- **One quarter for the banks, one observation for the macro series.** The bank workbook holds sixteen quarters each and the macro workbook holds 13,905 observations. What was compared is the most recent of each — 778 figures, not the whole history. The Case-Shiller set was additionally checked a second month.
- **Raw inputs, not the ratios built on them.** Forty-eight of each bank's lines are dollar amounts read straight off the form, and two are ratios the bank files itself. The FDIC's computed ratios, and the dashboard's z-scores, bands and alert logic, are arithmetic over figures this proves — and are not themselves checked here. That was the instruction: the base data.
- **Twenty of the Case-Shiller series are tied through a ratio of two cells**, not a level, because S&P does not publish adjusted levels free. That pins the month-on-month move, not the absolute level.
- **The provenance map passed rather than was proved universal.** Every code it cites was found on every one of these twelve filings and every number matched. A citation that is right for these filers and wrong for a bank filing a different form would not show up here.
- **Nothing here was checked by a second person.**